

Soln: $NJ(x)$: x lives in New Jersey $M(x)$: x lives within 50 miles of the ocean $S(x)$: x has seen the ocean

Domain: Set of all people in the world.

Premise: 1) $\forall x(NJ(x) \Rightarrow M(x))$ 2) $\exists x(\text{~~some~~ } NJ(x) \wedge \neg S(x))$

Conclusion: $\exists x(M(x) \wedge \neg S(x))$

Steps	Reason
1. $\forall x(NJ(x) \Rightarrow M(x))$	Premise
2. $\exists x(NJ(x) \wedge \neg S(x))$	Premise
3. $NJ(c) \wedge \neg S(c)$ for some c	Existential instantiation on 2
4. $NJ(c)$	Simplification on 3
5. $\neg S(c) \wedge NJ(c)$	Commutative law on 3
6. $\neg S(c)$	Simplification on 5
7. $NJ(c) \Rightarrow M(c)$	Universal instantiation on 1
8. $M(c)$	Modus ponens on 4, 7
9. $M(c) \wedge \neg S(c)$	Conjunction on 8, 6
10. $\exists x(M(x) \wedge \neg S(x))$	Existential generalization on 9

14) For each of these arguments, explain which rules of inference are used for each step.

a) "Linda, a student in this class, owns a red convertible. Everyone who owns a red convertible has gotten at least one speeding ticket. Therefore, someone in this class gotten a speeding ticket."

Soln: $S(x)$: x is a student in this class. $RC(x)$: x owns a red convertible
 $T(x)$: x has gotten at least one speeding ticket.

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